

Md Jueal Mia

PH.D. CANDIDATE · COMPUTER SCIENCE

Knight Foundation School of Computing and Information Sciences (KFSCIS), Florida International University, Miami, FL-33199

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Research Interests

Trustworthy AI, AI Security, Foundation Models, Large Language Models, Agentic AI, Federated Learning, Privacy-Preserving Machine Learning

Selected Highlights

- Published **7 peer-reviewed publications** (6 first-author; 3 journal articles and 4 conference papers), with **5 additional manuscripts under review** (4 journal articles and 1 conference papers).
- Submitted **3 U.S. patent applications** in trustworthy AI, federated learning, and foundation models.
- Contributed to multiple successful externally funded research proposals and projects, including **USDOT** and **NAIRR** initiatives in trustworthy AI, cybersecurity, and intelligent transportation systems.
- Recipient of the **Best Paper Award** at the *Third International Conference on Smart Systems: Innovations in Computing (SSIC 2021)*.

Education

Florida International University

Miami, FL, USA

PH.D. IN COMPUTER SCIENCE

Jan 2023 – Present

- CGPA: 4.00/4.00

Jahangirnagar University

Dhaka, Bangladesh

M.SC. IN COMPUTER SCIENCE AND ENGINEERING

2014 – 2015

Jahangirnagar University

Dhaka, Bangladesh

B.SC. IN COMPUTER SCIENCE AND ENGINEERING

2010 – 2014

Professional Experience

Department of Computer Science, Florida International University

Miami, FL, USA

GRADUATE RESEARCH ASSISTANT

May 2024 – Present

Department of Computer Science, Florida International University

Miami, FL, USA

GRADUATE TEACHING ASSISTANT

Jan 2023 – Apr 2024

Department of Computer Science, University of Nevada, Reno

Reno, NV, USA

GRADUATE RESEARCH ASSISTANT

Aug 2022 – Dec 2022

Department of CSE, Daffodil International University

Dhaka, Bangladesh

LECTURER (SENIOR SCALE)

Feb 2020 – Jan 2022

Department of CSE, Daffodil International University

Dhaka, Bangladesh

LECTURER

May 2016 – Jan 2020

Teaching Experience

Break Through Tech AI, Cornell Tech

New York, NY, USA

COURSE SUPPORT – MACHINE LEARNING FOUNDATIONS PROGRAM

May 2026 – Present

- Support instruction for the Machine Learning Foundations Program, helping students develop strong foundations in machine learning and artificial intelligence.
- Mentor students through project-based learning, coding exercises, office hours, and technical discussions.
- Evaluate programming assignments, quizzes, and projects while providing constructive feedback to improve student learning outcomes.
- Provide technical guidance on machine learning algorithms, model evaluation, and responsible AI practices in a collaborative learning environment.

Florida International University

Miami, FL, USA

GRADUATE TEACHING ASSISTANT

Jan 2023 – Apr 2024

- Supported instruction for undergraduate courses in Systems Programming and Java Programming.
- Mentored more than 10 undergraduate students through coursework, programming assignments, and technical projects.
- Mentored two NSF REU research interns on projects in trustworthy AI, federated learning, and secure machine learning.
- Facilitated hands-on learning activities focused on AI security, privacy-preserving machine learning, and large language models.

Daffodil International University

Dhaka, Bangladesh

ASSISTANT PROFESSOR

Feb 2022 – Jul 2022

- Taught undergraduate courses, provided academic advising, and supervised technical research projects.
- Supervised more than 50 undergraduate capstone and research projects spanning artificial intelligence, machine learning, and mobile application development.

Daffodil International University

Dhaka, Bangladesh

LECTURER (SENIOR SCALE)

May 2016 – Jan 2022

- Taught undergraduate courses to more than 500 students in Analysis of Algorithms, Software Engineering, Java, Python, and Machine Learning.
- Designed and developed the undergraduate curriculum for Analysis of Algorithms.
- Designed and developed the undergraduate curriculum for Mobile Application Development.
- Provided academic advising and mentored undergraduate students on technical projects and research.

Publications

JOURNAL PUBLICATIONS

1. **Md Jueal Mia**; M. Hadi Amini. (2025). QuanCrypt-FL: Quantized Homomorphic Encryption with Pruning for Secure Federated Learning. *IEEE Transactions on Artificial Intelligence*. doi: 10.1109/TAI.2025.3612906.
2. **Md Jueal Mia**; M. Hadi Amini. (2025). BART-FL: A Backdoor Attack-Resilient Federated Aggregation Technique for Cross-Silo Applications. *IEEE Transactions on Machine Learning in Communications and Networking*, vol. 3, pp. 1311–1325. doi: 10.1109/TMLCN.2025.3611398.
3. **Md Jueal Mia**; M. Hadi Amini. (2024). A Secure Object Detection Technique for Intelligent Transportation Systems. *IEEE Open Journal of Intelligent Transportation Systems*, vol. 5, pp. 495–508.

CONFERENCE PUBLICATIONS

1. **Md Jueal Mia**; M. Hadi Amini. (2025). An Empirical Analysis of Secure Federated Learning for Autonomous Vehicle Applications. In *Proceedings of Computing in Civil Engineering 2024*, pp. 417–426.
2. **Md Jueal Mia**; M. Hadi Amini. (2025). JaiLIP: Jailbreaking Vision-Language Models via Loss Guided Image Perturbation. In *Proceedings of the 2025 IEEE International Conference on Machine Learning and Applications (ICMLA)*, Boca Raton, FL, USA, pp. 1362–1368. doi: 10.1109/ICMLA66185.2025.00208.

3. Badhan Chandra Das; Md Tasnim Jawad; **Md Jueal Mia**; M. Hadi Amini; Yanzhao Wu. (2025). Jailbreaking Large Vision-Language Models in Intelligent Transportation Systems. In *Proceedings of the 2025 IEEE International Conference on Machine Learning and Applications (ICMLA)*, Boca Raton, FL, USA, pp. 1423–1429. doi: 10.1109/ICMLA66185.2025.00217.

ACCEPTED CONFERENCE PAPERS

1. **Md Jueal Mia**, M. Hadi Amini. Don't Overthink, Don't Underthink: Toward Adaptive Reasoning in Agentic AI. Accepted at *ACM AI Summit 2026*.

JOURNAL ARTICLES UNDER REVIEW

1. **Md Jueal Mia**; M. Hadi Amini. (2025). FedShield-LLM: A Secure and Scalable Federated Fine-Tuned Large Language Model. Submitted to *IEEE Transactions on Dependable and Secure Computing (TDSC)*.
2. M. Hadi Amini; **Md Jueal Mia**; Yasaman Saadati; Ahmed Imteaj; Seyedsina Nabavirazavi; Urmish Thakker; Md Zarif Hossain; Awal Ahmed Fime; S. S. Iyengar. (2025). Distributed LLMs and Multimodal Large Language Models: A Survey on Advances, Challenges, and Future Directions. *arXiv:2503.16585*. arXiv.
3. Hojat Allah Salehi; **Md Jueal Mia**; S. Sandeep Pradhan; M. Hadi Amini; Farhad Shirani. (2024). CorBin-FL: A Differentially Private Federated Learning Mechanism using Common Randomness. Submitted to *IEEE Open Journal of the Communications Society*.
4. Ahmed Aljohani; **Md Jueal Mia**; M. Hadi Amini. Backdoor-Resilient Lane Detection for Autonomous Vehicles: Assessment and Practical Defenses. Submitted to *IEEE Transactions on Intelligent Transportation Systems*.

CONFERENCE PAPERS UNDER REVIEW

1. **Md Jueal Mia**, Joaquin Molto, Yanzhao Wu, M. Hadi Amini. GUARD-SLM: Token Activation-Based Defense Against Jailbreak Attacks for Small Language Models. Submitted to *ACL ARR 2026*.

PATENTS

1. Mohammadhadi Amini, **Md Jueal Mia**. *A Privacy-Preserving Federated Fine-Tuned Large Language Model*. Submitted to the U.S. Patent and Trademark Office (USPTO), Pending, Serial No. 19/272,726, 2025.
2. Mohammadhadi Amini, **Md Jueal Mia**. *Systems and Methods for Adversarial and Jailbreak Attack-Resilient Federated Fine-Tuned Multimodal Large Language Models*. Patent Application Under Review.
3. Mohammadhadi Amini, **Md Jueal Mia**, Yanzhao Wu. *Systems and Methods for Defense against Jailbreak Attacks in Small Language Models*. Patent Application Under Review.

Presentations

CONTRIBUTED PRESENTATIONS

- **Md Jueal Mia** and M. Hadi Amini. "A Privacy-aware Cyber-attack-resilient Image Recognition Algorithm for Autonomous Vehicles." Poster presentation at the National Center for Transportation Cybersecurity and Resiliency (TraCR) Annual Conference, Greenville, SC, USA, May 6, 2024.

Research Funding & Proposal Development

COLLABORATIVE PROPOSAL CONTRIBUTIONS

- Contributed to the development of the successful USDOT National Center for Transportation Cybersecurity and Resiliency (TraCR) proposal, "Secure and Privacy-Preserving Federated Learning for Connected and Automated Vehicles."
- Contributed to the proposal development and technical research for "Empirical Security Analysis and Lightweight Defense Mechanisms Against Jailbreak Attacks in Small Language Models on Edge Devices", funded through the

National Artificial Intelligence Research Resource (NAIRR) Pilot (Grant No. NAIRR260008), supporting **3,000 Purdue Anvil GPU hours**.

- Contributed to the proposal development and technical research for *"Empirical Security Analysis and Defense Mechanisms Against Jailbreak Attacks in Large Language Models"*, funded through the National Artificial Intelligence Research Resource (NAIRR) Pilot (Grant No. NAIRR260027), supporting **\$4,500 in CloudBank credits**.

RESEARCH GRANTS AND STUDENT MENTORING

- Co-authored an NSF REU proposal, *"Enhancing Security Issues in Federated Learning for Resource-Constrained IoT Device Applications,"* and co-supervised undergraduate researcher Hannan Shah.
- Co-authored an NSF REU proposal, *"Enhancing Security Issues in Machine Learning for Application Development,"* and co-supervised undergraduate researcher Isabella Peng.

Technical Skills

- **Programming Languages:** Python, C, C++, Java, C#, R.
- **Generative AI and Foundation Models:** LLMs, SLMs, Agentic AI, VLMs, RAG, Prompt Engineering, Fine-Tuning (LoRA/PEFT), Hugging Face Transformers, LangChain, LangGraph, vLLM.
- **Machine Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib.
- **Tools and Software:** Linux, Git/GitHub, LaTeX (Overleaf), Jupyter Notebook, VS Code, PyCharm.

Research News Coverage

- **FIU News.** *"FIU Researchers Reveal How Altered Images Can Bypass AI Safeguards."* July 2026.
- **USA Today.** *"You can use altered images to trick AI systems, Florida study shows"* July 2026.
- **Yahoo Finance.** *"Florida International University Researchers Reveal How Altered Images Can Bypass AI Safeguards."* July 2026.
- **Sun Sentinel.** *"FIU Researchers Find New Weakness in AI Chatbots as Lawsuits Grow."* July 2026.

Selected Research and Project Collaborators

Florida International University (FIU): M. Hadi Amini (Associate Professor), Kemal Akkaya (Professor), Farhad Shirani (Assistant Professor), S. S. Iyengar (Distinguished University Professor), Yanzhao Wu (Assistant Professor), and Selcuk Uluagac (Eminent Scholar Chaired Professor).

Purdue University: Satish Ukkusuri (Professor) and Yiheng Feng (Assistant Professor).

University of Alabama: Mizanur Rahman (Associate Professor).

Southern Illinois University Carbondale: Ahmed Imteaj (Assistant Professor).

Morgan State University: Mansoureh Jeihani (Professor).

SambaNova Systems / University of Wisconsin: Urmish Thakker (Deep Learning Researcher).

Professional Service

EDITORIAL AND REVIEW ACTIVITIES

Journal Reviewer

- *IEEE Transactions on Information Forensics and Security (TIFS)* — **Impact Factor: 6.3**

- *IEEE Transactions on Dependable and Secure Computing (TDSC)* — **Impact Factor: 6.8**
- *IEEE Transactions on Vehicular Technology (TVT)* — **Impact Factor: 6.1**
- *IEEE Transactions on Consumer Electronics (TCE)* — **Impact Factor: 4.3**
- *IEEE Transactions on Machine Learning in Communications and Networking (TMLCN)*
- *Springer Nature Journals*

Conference Reviewer

- IEEE International Conference on Distributed Computing Systems (ICDCS 2024)
- International Conference on Machine Intelligence and Data Science Applications (MIDAS 2021)
- International Joint Conference on Advances in Computational Intelligence (IJCACI 2020)

SERVICE

- Program Committee Member, *IEEE International Conference on Tools with Artificial Intelligence (ICTAI)*, 2026.
- Organizing Committee Member, *International Joint Conference on Advances in Computational Intelligence (IJCACI)*, 2020.
- Organizing Committee Member, *International Collegiate Programming Contest (ICPC)*, 2018.

PROFESSIONAL DEVELOPMENT

- Faculty Research Training, *Road to Research*, Daffodil International University, 2018.
- Faculty Workshop on *Machine Learning and AI*, Daffodil International University, 2018.
- Faculty Workshop on *Research Proposal Development*, Daffodil International University, 2017.
- Participatory Engaging Techniques (PET) Teacher Training, Daffodil International University, 2017.

Awards, Fellowships, & Grants

- **Publication Grant** Recipient for Journal from the University Graduate School, Florida International University (2024, 2025).
- **Travel Grant** Recipient for Conference from the University Graduate School and Knight Foundation School of Computing and Information Sciences (KFSCIS), Florida International University (2024, 2025).
- **Best Paper Award** at the Third International Conference on Smart Systems: Innovations in Computing (SSIC), Springer (2021).
- In Recognition of Scholarly Publication in Reputed Indexed Journals for the year by Daffodil International University. (2020, 2021, 2022)
- Received **Best Performer-2018** by the Department of Computer Science and Engineering, Daffodil International University, for outstanding academic and professional contributions (2018).
- University Scholarship for Outstanding Academic Performance, awarded annually for top academic performance (2010–2014).
- Undergraduate Entrance Exam Scholarship for achieving the top position in the university entrance exam (2010–2011).